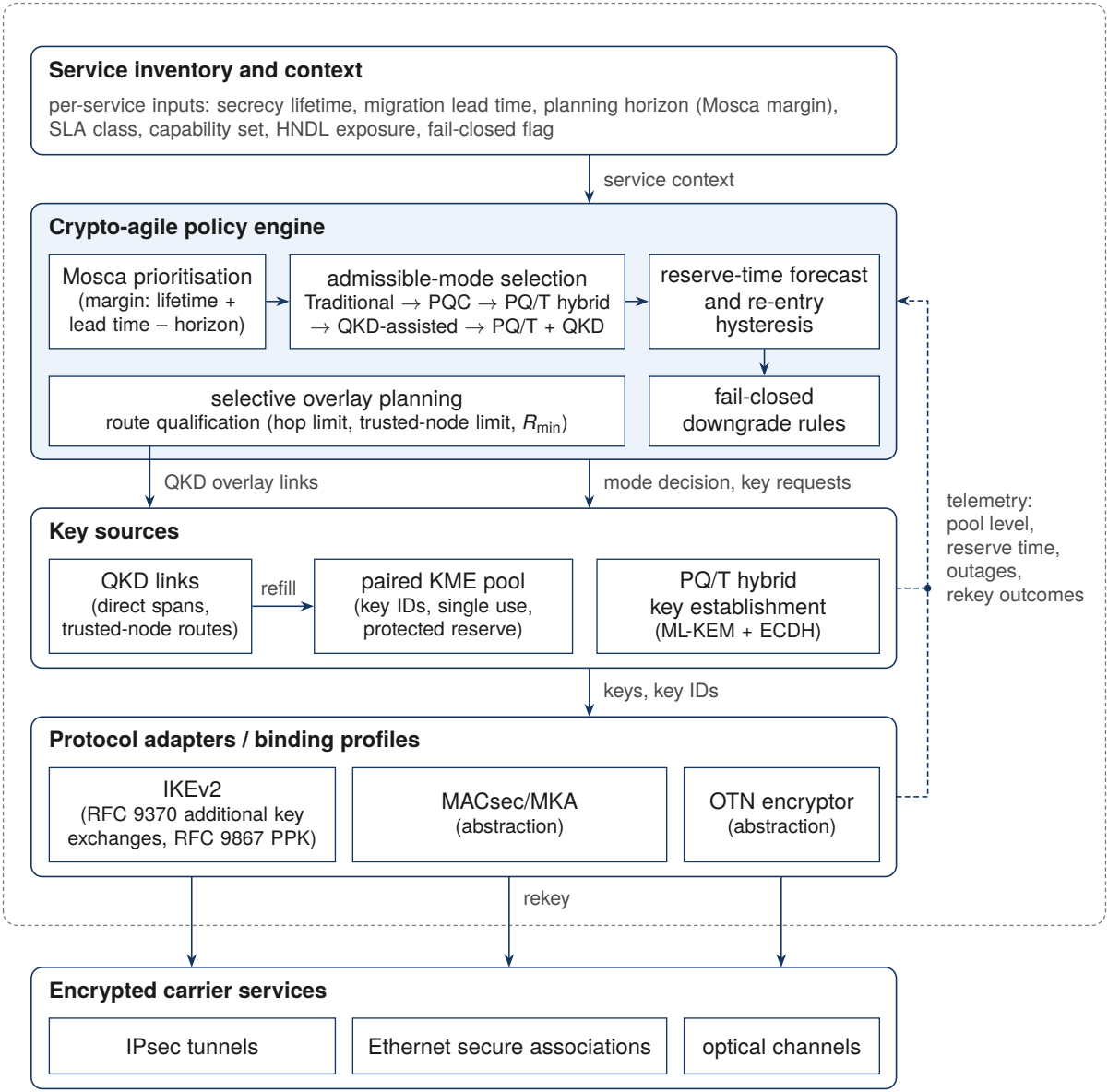

Service inventory and context
per-service inputs: secrecy lifetime, migration lead time, planning horizon (Mosca margin), SLA class, capability set, HNDL exposure, fail-closed flag
service context
Crypto-agile policy engine
Mosca prioritisation (margin: lifetime + lead time – horizon)
admissible-mode selection Traditional → PQC → PQ/T hybrid → QKD-assisted → PQ/T + QKD
reserve-time forecast and re-entry hysteresis
selective overlay planning route qualification (hop limit, trusted-node limit, R_{min})
fail-closed downgrade rules
QKD overlay links
mode decision, key requests
telemetry: pool level, reserve time, outages, rekey outcomes
Key sources
QKD links (direct spans, trusted-node routes)
refill
paired KME pool (key IDs, single use, protected reserve)
PQ/T hybrid key establishment (ML-KEM + ECDH)
keys, key IDs
Protocol adapters / binding profiles
IKEv2 (RFC 9370 additional key exchanges, RFC 9867 PPK)
MACsec/MKA (abstraction)
OTN encryptor (abstraction)
rekey
Encrypted carrier services
IPsec tunnels
Ethernet secure associations
optical channels